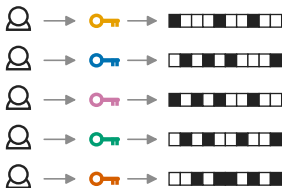


Fresh keys

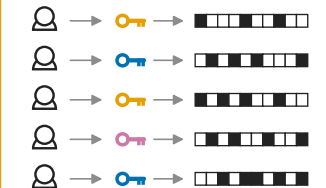
a new key for every record



covered by Theorem 1

Recurring pool of k keys

slot label hidden from attacker

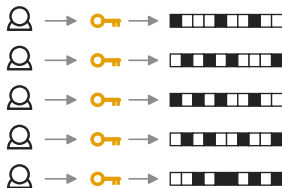


same colour = same hidden transform

empirical boundary

Single shared key

k = 1



stolen-token analogue

$$I(Y; T_1, \dots, T_n) = 0$$

top-1 at chance for every n

1 record at chance, 10 records not

leakage falls as k grows

transform is learnable

70-80 % top-1

Protocol identity-disjoint train / val / test · test keys never seen in training · one held-out gallery image per identity
8 nested record sets per identity · 3 model seeds · identity-clustered 95 % intervals · preregistered pass / fail per condition